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$$z^2 = (1 - i)(z - 2)$$

$$0 = -z^2 + z - 2 - iz + 2i$$

$$0 = -z^2 + (1 - i)z + (i - 1)2$$

$$D = -2i + 8i - 8 = 6i - 8$$

De vierkantswortel van $6i - 8$:

$$6i - 8 = x^2 - y^2 + 2xyi$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = -8 \\ x^2(-y)^2 = -9 \end{cases}$$

$$\lambda^2 + 8\lambda - 9 = 0$$

$$D = 94 + 36 = 100$$

$$\lambda_1 = \frac{-8 + 10}{2} = \frac{-18}{2} = -9 \Rightarrow y = \pm 3$$

$$\lambda_2 = 1 \Rightarrow x = \pm 1$$

$$x + yi \in \{1 + 3i, -1 - 3i\}$$

$$\text{Dus uit } 0 = -z^2 + (1 - i)z + (i - 1)2$$

$$z_1 = \frac{-1 + i + (1 + 3i)}{-2} = -2i$$

$$z_2 = \frac{-1 + i - (1 + 3i)}{-2} = i + 1$$

References